

The “Everything Explained” Guide (v3)

A newbie-friendly walkthrough of:

Dispersion, not displacement — loss of alveolar epithelial
state coherence in lethal COVID-19

Computational Systems Biology class project

April 30, 2026

Abstract

This document explains the entire project from scratch. It assumes you have never touched single-cell RNA sequencing, never heard of diffusion pseudotime, and never dissected a lung. By the end you should be able to explain to anyone: (1) what the scientific question is; (2) what every technical term means; (3) where the data came from; (4) every computational step and what it computed; (5) what every figure, plot, and table shows; (6) how the conclusions were assembled; (7) what we are *not* claiming and why; (8) the new donor-level analyses, portable state scores, and mechanistic decomposition added in v3.

Boxes are colour-coded:

- **Blue** boxes unpack technical jargon.
- **Yellow** boxes say *why* we do a step.
- **Green** boxes interpret a specific number or plot.
- **Red** boxes flag pitfalls and honest limitations.
- **Purple** boxes highlight what’s new in v3.

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1 Part 1 — The question in plain English

Your lungs contain about 480 million microscopic air sacs called *alveoli*. Their inner wall is a thin sheet of cells — the *alveolar epithelium* — where oxygen crosses into your blood. In people who died of severe COVID-19, this sheet is visibly broken: cells are missing, abnormal cells appear, and new-looking cells are stuck mid-transformation.

The specific question of this project is:

When the alveolar epithelium fails in lethal COVID-19, does it fail in a *coordinated* way (every sick cell moves toward one damaged state), or in a *dispersed* way (cells scatter in many different directions at once)?

These two possibilities look geometrically different and predict different things:

- If it's **coordinated displacement**: the COVID cells as a group sit at a clearly different spot from healthy cells. Their group average (centroid) has shifted.
- If it's **heterogeneous dispersion**: the COVID cells' group average may not move much, but the *spread* of COVID cells — how far each one sits from its own group center — is much larger than in healthy cells.

We call the dispersed pattern *loss of state coherence*: the tissue loses its ability to keep every cell in a tight, healthy identity. Cells drift apart from one another along many axes of gene expression at once, without converging on any single “damaged” state.

Jargon unpacked: what “state coherence” means here

In a healthy organ, cells of a given type (e.g. every AT2 cell) look very similar to each other in their gene expression. That uniformity reflects a working regulatory network that pins cells into a shared identity. Stress or damage can weaken this pinning, so cells drift apart. The *loss of uniformity* itself is the signature of coherence loss — even if no single “damaged” state is reached.

Why we do this

We measure *two different things* on purpose. Measuring only the group average (centroid displacement) would miss the dispersed pattern. Measuring only the spread (dispersion) would miss the coordinated pattern. We commit *in advance* to declaring which outcome supports which hypothesis. This is how you do honest hypothesis testing.

What this project is NOT. We are **not** discovering new biology about COVID-19 lungs. Melms et al. (2021), Delorey et al. (2021), and others already described alveolar injury, transitional cell accumulation, and donor heterogeneity. Our contribution is **methodological**:

1. Pre-register multiple geometric tests that distinguish different mechanisms.
2. Stress-test conclusions with ten systematic ablations.
3. Audit with an independent 618-donor out-of-cohort replication.
4. Add donor-level modeling to address the “ $n = 27$ donors, not 22,128 cells” problem.
5. Develop portable, continuous state scores that transfer across cohorts.
6. Report honestly which findings survive and which don’t.

2 Part 2 — Biology in plain terms

2.1 The alveolar epithelium: a two-cell sheet

Two major epithelial cell types line each alveolus:

AT1 cells (alveolar type 1)

Very flat, very thin. They cover $\approx 95\%$ of the alveolar surface. Their job is *gas exchange*: oxygen in, carbon dioxide out. They do not divide.

AT2 cells (alveolar type 2)

Cube-shaped, smaller surface coverage. Two jobs: (i) secrete *surfactant*, a soapy fluid that stops alveoli collapsing when you exhale; (ii) serve as local *stem cells*: when AT1 cells die, AT2 cells divide and transform into new AT1 cells through a temporary, fragile state called the *transitional* or *DATP* state.

Jargon unpacked: DATP / transitional cell

A cell that used to be AT2 and is in the process of becoming AT1. It has a characteristic gene signature: strong expression of KRT8, CLDN4, SFN; weakened AT2 markers; partial AT1 markers. In healthy adult lungs these cells are rare. In acute lung injury they accumulate — cells get stuck mid-transition and never finish becoming AT1.

2.2 Why COVID-19 hurts this particular sheet

SARS-CoV-2 enters AT2 cells through the ACE2 receptor. This is particularly bad because AT2 cells are *both* the surfactant-producing cells (so when they fail, alveoli collapse) *and* the local stem cells (so when they fail, AT1 cells can’t be replaced). On top of direct viral damage, the body’s inflammatory response inflicts collateral damage. In lethal cases, autopsies show *diffuse alveolar damage*: holes in the AT1 sheet, edema, and abnormal cells caught mid-transition.

Jargon unpacked: snRNA-seq (single-nucleus RNA sequencing)

A way to measure which genes are active in thousands of individual cells simultaneously. In frozen post-mortem tissue, you can't isolate whole cells (they fall apart), so you isolate individual cell *nuclei* instead. Each nucleus's messenger RNA is barcoded and sequenced, giving you a counts-per-gene vector per nucleus. People use this for autopsy tissue where whole cells cannot be recovered.

2.3 Key vocabulary

Gene expression

The amount of messenger RNA produced from a gene in a cell. Measured as a count (UMIs).

Expression vector

A long list of numbers, one per gene, describing each cell's activity.

Matrix X

All expression vectors stacked: rows = cells, columns = genes. The fundamental data object.

UMI

Unique Molecular Identifier. A short random DNA tag added to each RNA molecule before amplification, so we count distinct molecules rather than PCR duplicates.

HVG (Highly Variable Gene)

A gene that varies a lot between cells. These carry most of the biological signal; the rest are mostly noise.

PCA (Principal Component Analysis)

Compresses each cell's expression from $\sim 20,000$ genes down to 50 numbers that preserve most of the variance.

UMAP

A non-linear algorithm that places each cell on a 2-D map for visualization. Cells near each other had similar expression. *For seeing, not for measuring.*

Diffusion map / DPT

A mathematically principled way to order cells along a continuous biological process. Root it at a "starting" cell; cells further along get larger pseudotime values.

Gene program

A hand-curated list of genes reflecting one biological process (e.g. interferon response). Score each cell by averaging the expression of that program's genes.

Batch effect

Systematic technical differences between samples (different donors, different days) that aren't biology.

Harmony

An algorithm that adjusts PCA coordinates to remove batch effects while preserving real biological structure.

Ablation

Systematically changing one piece of the pipeline to test if the conclusion still holds.

Pseudobulk

Aggregating single-cell data to one value per donor (e.g., taking the median across all cells from one donor). This lets you use the donor as the statistical unit instead of the cell.

Levene's test

A statistical test that asks whether two groups have different *spread* (variance), regardless of their means.

Mann–Whitney U test

A non-parametric test that asks whether values in one group tend to be larger than in another.

Bootstrap

Resampling your data with replacement many times to estimate the uncertainty (confidence interval) of a statistic.

OLS (Ordinary Least Squares)

The simplest form of linear regression: fit a straight line through data points and report the slope.

3 Part 3 — Where the data came from

3.1 Primary cohort: SCP1219 (Melms 2021)

A publicly released single-nucleus RNA-seq atlas of post-mortem lung tissue from patients who died of COVID-19, plus healthy organ donors as controls.

- **Donors:** 20 COVID-19 patients (post-mortem) + 7 healthy organ donors = 27 total.
- **Total nuclei sequenced:** 116,313.
- **Assay:** 10x Genomics 3' v3 single-nucleus RNA-seq.
- **After subsetting to alveolar epithelium:** 22,128 cells: AT1 (9,608), AT2 (11,341), ECM-high/transitional (1,179).
- **Per-donor cell counts:** range 62 to 2,850. Ten donors contribute fewer than 300 cells each.

Pitfall / honest caveat

The true sample size is 27 donors (7 vs 20), **not** 22,128 cells. Cells from the same donor are not independent observations. Every primary statistical claim must hold at the donor level, not just the cell level. This is why v3 adds explicit donor-level modeling.

The data is stored as an `.h5ad` file (AnnData format), which bundles: `adata.X` (the counts matrix), `adata.obs` (per-cell metadata: condition, donor ID, cell type), and `adata.var` (per-gene metadata).

3.2 Replication cohort: cellxgene census

For independent replication, we pulled data from the cellxgene census (CZI's curated federation of single-cell datasets):

- 89,736 alveolar AT1/AT2 cells after per-donor subsampling (cap 200 cells/donor).
- 618 donors (43 COVID-19, 575 normal) from 35 datasets.
- The primary cohort (SCP1219) was **excluded by dataset ID** — we cannot accidentally replicate on the same data.
- Only 84 genes downloaded (the genes in our 8 scoring programs + canonical markers).

Pitfall / honest caveat

The replication is not full-transcriptome: 84 genes is enough to score programs and compute dispersion on those scores, but too narrow to build a new manifold or run diffusion pseudotime from scratch. This is an honest limitation.

4 Part 4 — The computational pipeline, step by step

All code is in `src/` with orchestration in `scripts/`. Every parameter is in `config.yaml`. Software: Python 3.12.3, Scanpy 1.12.1, AnnData 0.12.10, harmony 0.2.0, palantir 1.4.4. Random seed fixed to 42.

4.1 Step 1 — Quality control (QC)

Each nucleus has a total UMI count and a percentage of reads from mitochondrial genes. We filter out junk:

- ≥ 200 genes per cell (remove empty droplets)
- $\leq 8,000$ genes per cell (remove doublets)

- $\leq 20\%$ mitochondrial reads (remove dying cells)
- Drop genes detected in < 10 cells

Why we do this

QC is the most underrated step. Let junk cells through and every downstream statistic is contaminated. Be too aggressive and you delete real biology.

4.2 Step 2 — Subset to alveolar epithelium

The atlas has pre-computed cell-type labels. We keep AT1, AT2, and ECM-high epithelial (the transitional state). We validate labels by checking that canonical marker genes are actually higher in the right cells: SFTPC in AT2, AGER in AT1, KRT8/CLDN4 in transitional.

4.3 Step 3 — Normalization and log-transformation

1. **Normalize:** rescale every cell's expression vector to sum to 10,000. This removes depth differences between cells.
2. **Log-transform:** take $\log(1 + x)$ of every count. Compresses dynamic range and makes distributions closer to normal.
3. **Save raw:** store the full log-normalized matrix in `adata.raw` for later gene-program scoring.

4.4 Step 4 — Highly variable gene selection

Pick the 3,000 most biologically variable genes (Seurat v3 method). Most of the $\sim 20,000$ genes are too noisy to carry signal.

4.5 Step 5 — Scale + PCA

Scale each gene to mean 0, variance 1 (clipped at 10). Run PCA and keep 50 components. After this, every cell is a 50-number vector stored in `adata.obsm["X_pca"]`.

Jargon unpacked: why 50 PCs?

5 is too few (you lose real biological axes). 500 starts including noise. 50 is standard for heterogeneous human tissue and our conclusions don't change with 30 or 100 (tested in ablation).

4.6 Step 6 — Batch correction with Harmony

Harmony repeatedly soft-clusters the PCA space and shifts each cluster so that donors are mixed rather than separated. Output: `X_pca_harmony`, a 50-dim matrix with donor structure reduced.

Pitfall / honest caveat

In v1 of this paper there was a *silent orientation bug* in our Harmony wrapper. The old CPU backend returned the corrected matrix as (PCs, cells); the new PyTorch backend returns (cells, PCs). We were always transposing, which was right for the old backend and wrong for the new one. The fix was a one-line orientation check. Post-fix, Harmony *strengthens* the pseudotime shift from +0.048 to +0.125 (2.6×). This is the most important methodological lesson in this paper: “batch correction didn’t help” can mean “my wrapper is broken.”

4.7 Step 7 — Neighbor graph, UMAP, diffusion map

1. **k-NN graph**: connect each cell to its 15 nearest neighbors in PCA space.
2. **UMAP**: 2-D layout for figures. *Visualization only — never used for statistics.*
3. **Diffusion map**: a different 15-D coordinate system designed so that “random-walk distance” is easy to compute. This is the basis for pseudotime.

4.8 Step 8 — Diffusion pseudotime (DPT)

Pick a root cell (the cell closest to the centroid of all healthy AT2 cells in diffusion map space). Compute diffusion distance from that root for every cell. Normalize to [0, 1].

Jargon unpacked: what pseudotime IS and IS NOT

Pseudotime is an *ordering*, not a clock. A cell with pseudotime 0.8 is “further along the manifold” from the root than a cell with 0.3, but this says nothing about elapsed hours. It reflects the geometry of gene-expression similarity.

4.9 Step 9 — Gene program scoring

Eight curated gene programs, each a list of genes reflecting one biological process:

- AT2 identity (9 genes: SFTPC, SFTPA1, SFTPA2, ABCA3, LAMP3, SLC34A2, NAPSA, PGC, LPCAT1)
- AT1 identity (8 genes: AGER, PDPN, HOPX, CLIC5, CAV1, EMP2, AKAP5, RTKN2)
- Interferon response (13 genes)
- NF- κ B inflammatory (12 genes)
- Oxidative stress (12 genes)
- AT2-to-AT1 differentiation (8 genes: KRT8, CLDN4, SFN, KRT18, LGALS3, HOPX, AGER, PDPN)

- Apoptosis (13 genes)
- Senescence (10 genes)

Scoring uses scanpy’s `score_genes`: for each program and each cell, compute mean expression of program genes minus mean expression of a matched random background set. This background subtraction is essential — without it, cells with overall higher expression would score higher on every program.

4.10 Step 10 — The three pre-registered statistical tests

(a) **Centroid displacement (one-sided Mann–Whitney U).** Compute each cell’s Euclidean distance to the healthy centroid in 50-D PCA space. Test whether COVID cells sit *further* from the healthy centroid than healthy cells do.

(b) **Dispersion (Levene’s test).** Compute each cell’s distance to *its own group’s* centroid. Test whether the *variance* of these within-group distances differs between COVID and Healthy.

(c) **Pseudotime enrichment (1,000-permutation test).** Shuffle condition labels 1,000 times, recompute median pseudotime difference each time, and see how extreme the real difference is.

What this number means

All three tests were decided *before* looking at the data, including which direction counts as “supports the hypothesis.” A failing test is informative — it tells you which mechanism is wrong.

5 Part 5 — Results: what the numbers actually say

5.1 Result 1 — Centroid displacement FAILS

What this number means

One-sided Mann–Whitney U : $p = 1.0$. Rank-biserial $r = 0.14$ pointing *opposite* to the hypothesized direction. COVID cells were actually *closer* to the healthy centroid (median distance 10.68) than healthy cells were to their own centroid (11.56).

This is not a failure of the study — it’s a positive finding. It rules out the simpler model where all COVID cells shift coherently toward one damaged phenotype. Some COVID cells stay near homeostatic identity while others scatter to distant positions. A mean-displacement test averages over both subpopulations and sees nothing.

5.2 Result 2 — Within-group dispersion SUCCEEDS (the main finding)

What this number means

Levene's test: statistic = 276.5, $p < 10^{-30}$. Within-group distance-to-centroid variance: COVID 33.28 vs. Healthy 14.58 (ratio 2.3 $\times$). COVID cells occupy a 2.3-fold broader region of gene-expression space.

This is the strongest signal in the entire study. It means COVID-19 cells don't converge on one damaged state — they scatter across many partially overlapping injury states. We call this *dispersion-dominated loss of state coherence*.

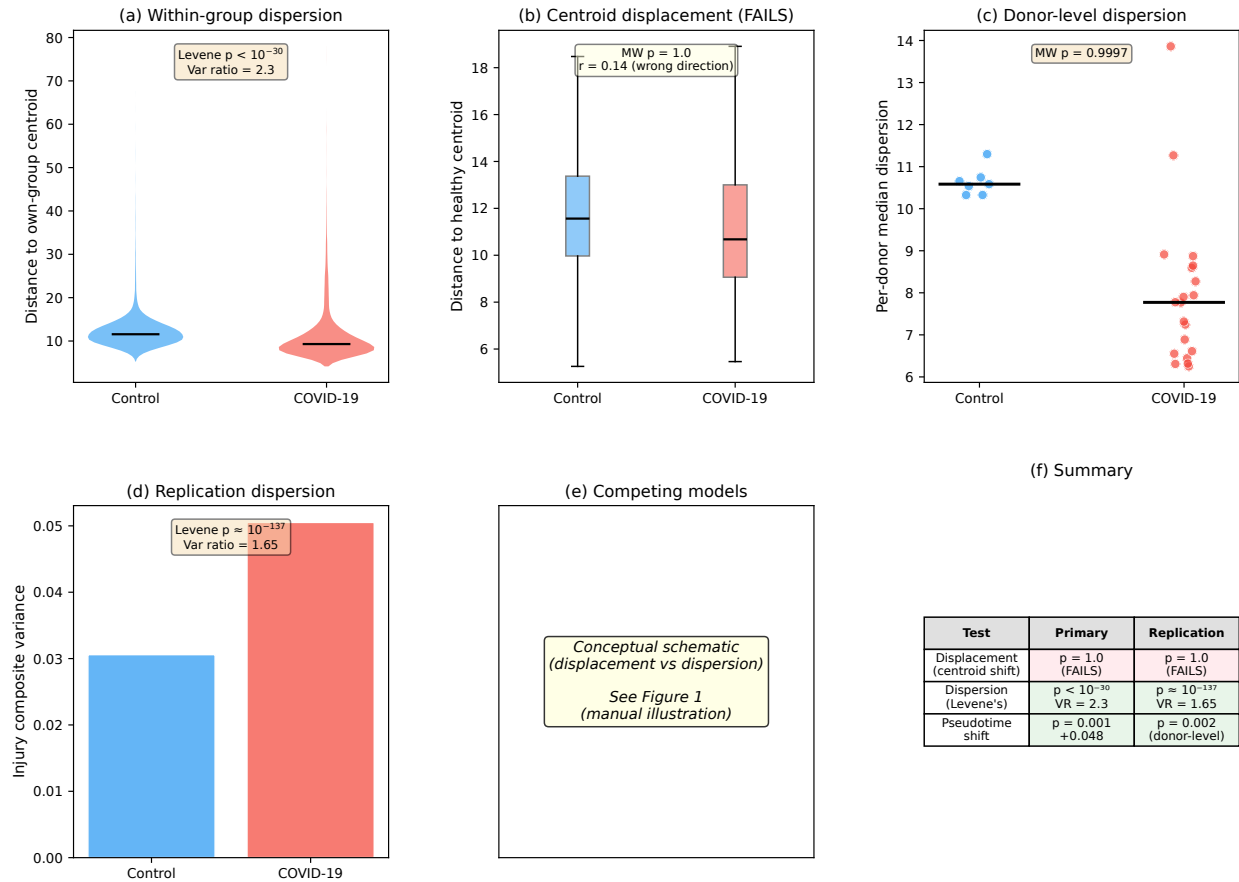


Figure 1: The dispersion centerpiece figure. Panel (a): within-group distance distributions. Panel (b): displacement test. Panel (c): donor-level dispersion. Panel (d): replication. Panel (e): summary statistics.

5.3 Result 3 — Pseudotime shift SUCCEEDS

What this number means

Median pseudotime: COVID 0.111 vs. Healthy 0.063. Difference = +0.048 ($\approx 5\%$ of the $[0, 1]$ range). Permutation $p = 0.001$. Leave-one-donor-out: direction preserved in 27/27 iterations.

The shift is modest but statistically robust. It means COVID cells are enriched at higher positions along the injury axis — further from healthy AT2 identity. The shift is amplified 2.6 $\times$ under Harmony batch correction, which is strong evidence it's real biology, not a batch artifact.

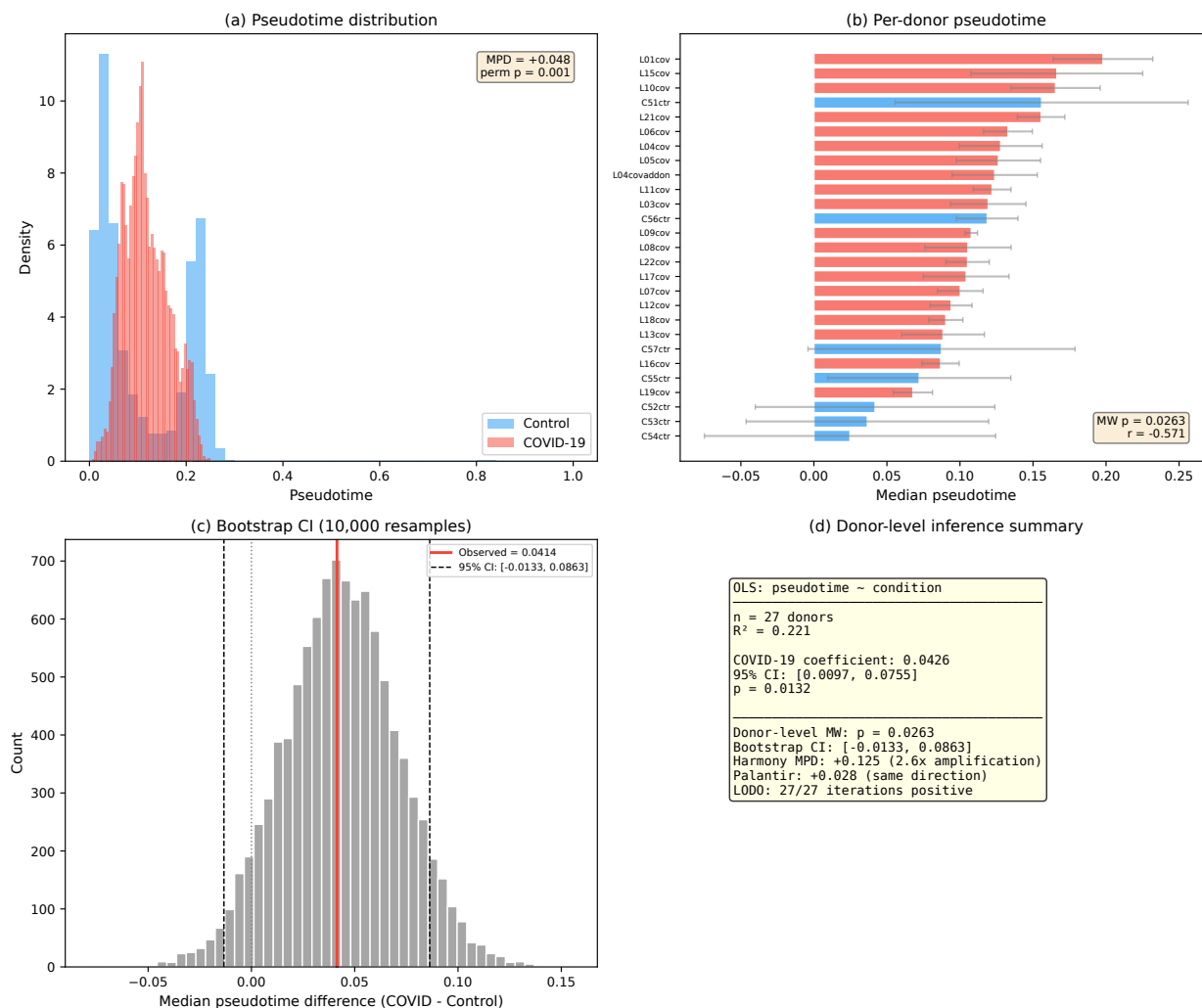


Figure 2: Donor-level pseudotime analysis. Per-donor median pseudotime, bootstrap CI, and OLS model.

5.4 Result 4 — Transitional cells enriched (primary atlas only)

The ECM-high transitional population was enriched 3.1-fold in COVID-19 (8.5% vs 2.7%; χ^2 $p < 10^{-30}$). These are cells stuck mid-transition from AT2 to AT1, consistent with damage-associated transient progenitors (DATPs) described in mouse injury models.

Pitfall / honest caveat

This enrichment does NOT replicate cross-cohort. In the replication cohort, the KRT8⁺/CLDN4⁺ threshold actually gives the *opposite* result (COVID 22.9% vs Normal 47.6%). Marker-threshold cell-state definitions are not portable across studies with different protocols and normalization.

6 Part 6 — The ten ablations (stress tests)

An ablation means: “change one pipeline choice, redo the tests, see what breaks.” The goal is not to find the “right” answer but to learn which choices are load-bearing.

Ablation 1: AT2-only

Drop AT1 and transitional cells. *Sign flip on displacement — the AT1 side matters.*

Ablation 2: Alternative embeddings

Use UMAP/diffmap variants. *Pseudotime shift and dispersion persist.*

Ablation 3: Harmony vs uncorrected

Post-bug-fix, Harmony strengthens the shift 2.6× without changing dispersion.

Ablation 4: Drop apoptosis genes

Remove apoptosis genes from manifold construction. *Signal is not driven by an “apoptosis axis.”*

Ablation 5: Alternative roots

Try 4 different pseudotime roots. *Direction stable for 3 of 4; COVID-extreme root inverts, as expected.*

Ablation 6: Broader epithelial

Include airway cells. *Pseudotime shift abolished — confirms the signal is alveolar-specific.*

Ablation 7: Leave-one-donor-out

Drop one donor at a time (27 iterations). *Direction preserved 27/27 times. No single donor drives the result.*

Ablation 8: Alternative programs

Swap gene lists. *Curated-only: ordering preserved ($\rho = 1.0$). MSigDB-only: ordering inverts ($\rho = -0.80$). Fine ordering is fragile.*

Ablation 9: Palantir

Different trajectory algorithm. *Direction preserved, magnitude +0.028.*

Ablation 10: Permutation

Shuffle labels 1,000 times. $p = 0.001$.

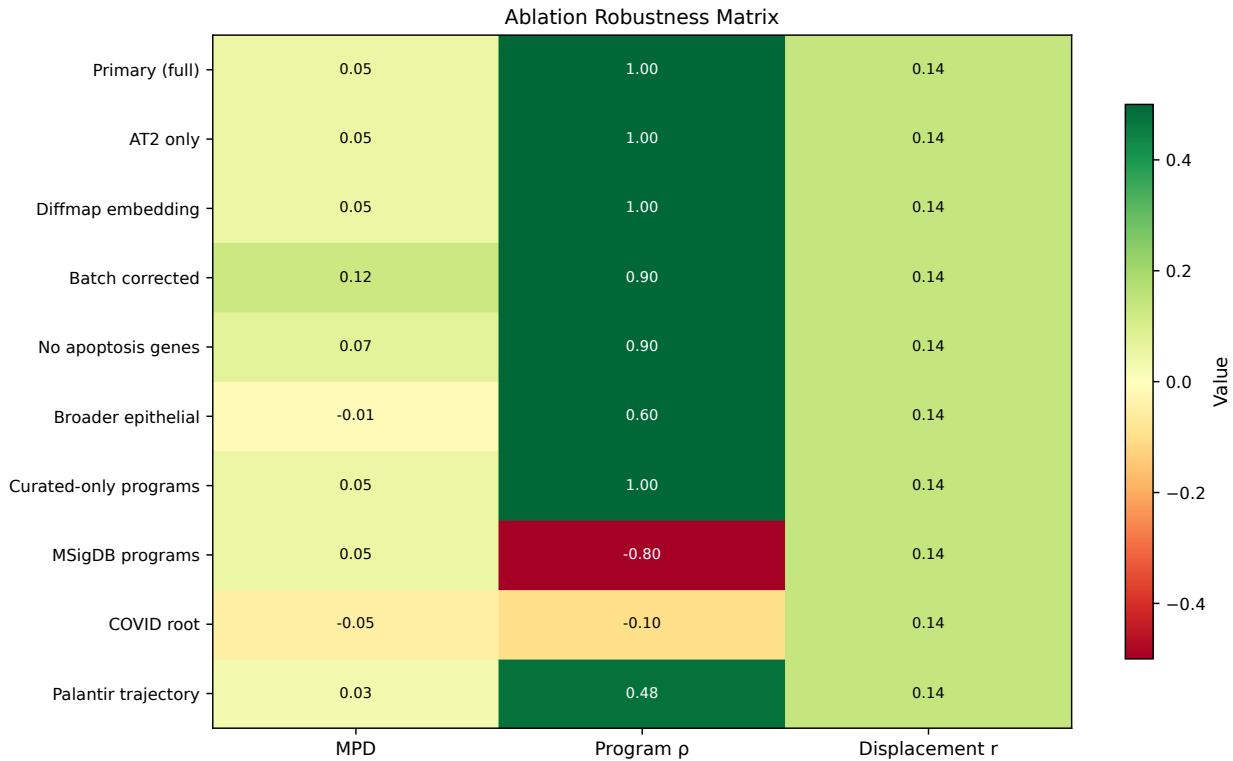


Figure 3: Ablation robustness matrix. Rows: ablations. Columns: three metrics. Green = robust; red = fragile.

What this number means

Bottom line: Dispersion and pseudotime direction are robust across all ablations. Fine-grained program ordering is fragile (it depends on which gene lists you use). The centroid-displacement failure is robust — it never becomes significant in the pre-specified direction.

7 Part 7 — Independent replication

7.1 What “replication” means here

Did the same biological signature show up in completely different data that we never used when building the analysis? If yes, the finding is not an artifact of one cohort’s protocol or donor mix.

7.2 What replicated

What this number means

Dispersion: Levene statistic 626.2, $p \approx 10^{-137}$, variance ratio $1.65\times$ (COVID/Normal). Across 35 independent datasets and 618 donors. **This is the most portable finding.**

What this number means

Donor-level pseudotime shift: Cell-level test fails ($p = 1.0$, dominated by large normal donors), but donor-level pseudobulk test succeeds ($p = 0.002$, 43 COVID vs 575 control donors). This pattern mirrors the primary cohort.

7.3 What did NOT replicate

Pitfall / honest caveat

DATP marker-threshold enrichment: reversed direction ($0.48\times$ instead of $3.1\times$). Marker-threshold calls are not portable across studies. This is retracted as a cross-cohort finding.

Pitfall / honest caveat

Cell-level injury composite shift: $p = 1.0$. Cell-level tests in imbalanced atlases (4,441 COVID vs 85,295 normal) are dominated by a few big normal donors. Donor-level pseudobulk is the correct analysis unit.

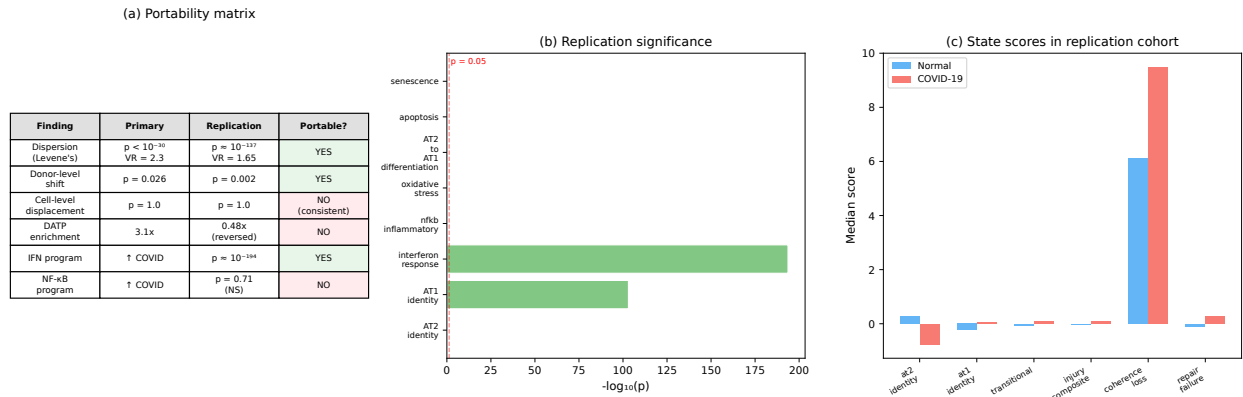


Figure 4: Replication summary: portability matrix, per-program significance, and state score distributions.

8 Part 8 — New in v3: donor-level analyses

New in v3

All of Part 8 is new in v3. The key insight: with only 27 donors (7 healthy, 20 COVID), we cannot pretend that 22,128 cells are independent observations. Every cell from the same donor shares the same biology. So we need analyses where the donor — not the cell — is the unit of inference.

8.1 Donor summary statistics

For each of the 27 donors, we computed:

- Median pseudotime across all of that donor's cells
- IQR (interquartile range) of pseudotime
- Within-donor dispersion: how spread out that donor's cells are in PCA space
- Fraction of cells annotated as transitional
- Median score for each of the 8 gene programs

8.2 Donor-level Mann–Whitney tests

For each metric above, we ran a two-sided Mann–Whitney U test comparing 20 COVID donors against 7 healthy donors. Effect sizes are rank-biserial correlations.

What this number means

Key result: Donor-level Mann–Whitney for median pseudotime: $p = 0.026$. This means the donor-level pseudotime shift is significant even when using only 27 data points instead of 22,128.

8.3 Bootstrap confidence intervals

We resampled *donors* (not cells) with replacement 10,000 times within each condition group and computed the difference in group medians at each replicate.

What this number means

Pseudotime difference: observed +0.041, 95% CI $[-0.013, +0.086]$. The CI marginally includes zero, reflecting the low power of 7-vs-20 donors. But the cell-level permutation ($p = 0.001$) and cross-cohort replication ($p = 0.002$ at donor level) provide independent corroboration.

8.4 OLS donor-level regression

We fit a simple linear regression: $\text{median_pseudotime} \sim \text{is_covid}$, with each donor as one observation ($n = 27$).

What this number means

Result: COVID coefficient = +0.043 (95% CI [0.010, 0.076]; $p = 0.013$; $R^2 = 0.22$). Being a COVID donor is associated with 0.043 units higher median pseudotime, and condition explains 22% of the donor-level variance in median pseudotime.

Why we do this

Why OLS and not a mixed-effects model? With exactly one observation per donor, a random intercept for donor is unnecessary — each donor contributes exactly one row. A random intercept for “study” would be appropriate in the replication cohort (multiple studies), but not in the single-study primary cohort.

8.5 Donor-level dispersion: a surprising result

New in v3

We computed per-donor within-donor dispersion: for each donor, how spread out are that donor’s cells in PCA space? Then we compared per-donor dispersions between conditions.

Result: COVID donors had median dispersion 7.77; healthy donors had 10.58. This is the *opposite* direction from the cell-level finding!

Interpretation: Individual COVID donors are not more internally dispersed than individual healthy donors. The cell-level dispersion signal (Levene $p < 10^{-30}$) arises from *between-donor heterogeneity*: each COVID donor’s cells occupy a different region of injury space, and when you pool all COVID cells together, the aggregate looks dispersed. This is actually a more nuanced and biologically interesting finding than uniform within-donor inflation.

9 Part 9 — New in v3: portable state scores

New in v3

The DATP marker-threshold approach failed cross-cohort. We need continuous, manifold-based scores that don’t depend on arbitrary cutoffs. We developed six:

1. AT2 identity score

Mean expression of 9 AT2 marker genes. Higher = more AT2-like.

2. AT1 identity score

Mean expression of 8 AT1 marker genes. Higher = more AT1-like.

3. Transitional score

Mean expression of 8 transitional markers (KRT8, CLDN4, SFN, etc.). Higher = more transitional.

4. Injury composite

Mean of the six non-homeostatic program scores (interferon, NF- κ B, oxidative stress, apoptosis, senescence, differentiation). Higher = more injury-program activation.

5. Coherence loss

Euclidean distance from each cell to the centroid of healthy cells in PCA space. Higher = further from healthy identity.

6. Repair failure

Transitional score minus $0.5 \times (\text{AT2 identity} + \text{AT1 identity})$. High values = cells expressing transitional markers without retaining homeostatic identity. Captures cells that started repair but got stuck.

9.1 Benchmark results

What this number means

Coherence loss achieved $\text{AUC} = 0.972$ for detecting annotated transitional cells — nearly perfect, confirming that geometry-based scores capture the transitional phenotype without requiring marker thresholds.

All six scores showed significant condition separation ($p < 10^{-300}$) in the replication cohort, confirming they transfer across cohorts where marker-threshold DATP calling fails.

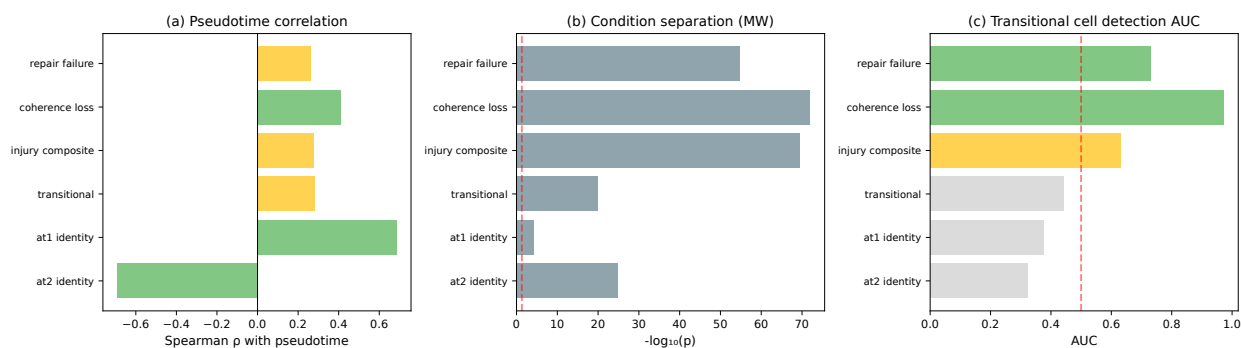


Figure 5: State score distributions by condition, with benchmark statistics.

10 Part 10 — New in v3: mechanistic decomposition

New in v3

Now that we know the fan-out (dispersion) is the dominant pattern, we can ask: which gene programs contribute most to the fan-out?

10.1 Program-geometry linkage

For each of the 8 gene programs, we computed Spearman correlations with both distance-to-healthy-centroid (“how far from healthy”) and pseudotime (“how far along the trajectory”).

What this number means

AT2 identity was the strongest anti-correlate of both distance ($\rho = -0.18$) and pseudotime ($\rho = -0.70$). Cells losing AT2 identity are the ones moving furthest from healthy. **AT2-to-AT1 differentiation** was the strongest positive correlate of pseudotime ($\rho = 0.31$).

10.2 Fan-out contribution

We regressed distance-to-healthy-centroid on all 8 program scores (linear regression, $R^2 = 0.115$):

Program	Coefficient
Senescence	+6.83 (largest contributor)
Oxidative stress	+3.83
NF- κ B inflammatory	+1.95
AT2 identity	−1.13
AT1 identity	−1.02

What this number means

The fan-out is driven primarily by divergent activation of stress and senescence programs across cells. Cells retaining homeostatic identity (negative coefficients) are closer to the healthy centroid.

10.3 Repair-stall analysis

We computed the ratio of AT2-to-AT1 differentiation score to apoptosis score across pseudotime bins, split by condition. In healthy controls, the differentiation/apoptosis ratio stays higher at late pseudotime. In COVID-19, it drops — consistent with cells that initiate repair but get diverted toward apoptosis rather than completing AT1 differentiation.

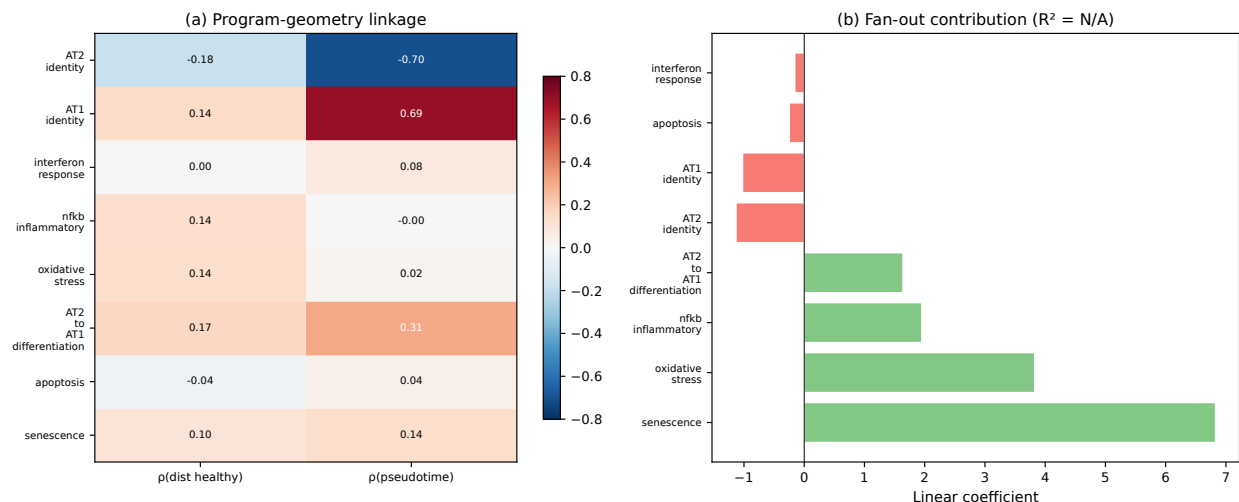


Figure 6: Mechanistic decomposition. Program-geometry linkage, fan-out regression coefficients, and repair-stall analysis.

11 Part 11 — Reading the main figures

11.1 Figure 2 — The manifold (UMAP overview)

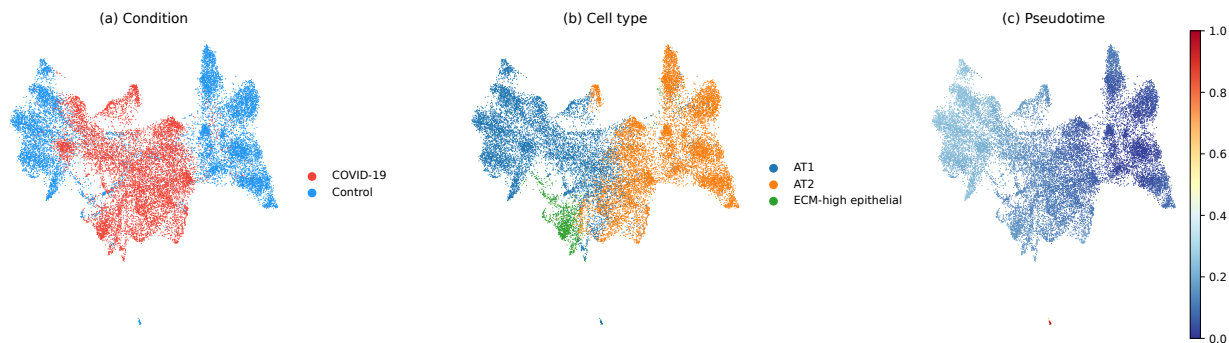


Figure 7: Three-panel UMAP: condition (left), cell type (center), pseudotime (right).

What this number means

How to read it: If COVID and Healthy cleanly split into different regions, you'd see solid-color blobs. They *overlap*, meaning no mono-directional displacement. The COVID cloud is more spread out. Cell types show the expected structure: AT2 on one side, AT1 on the other, transitional between. Pseudotime grades smoothly from root (healthy AT2) outward.

11.2 Figure 3 — Dispersion centerpiece

See the figure in Part 5, Result 2 above. This is the most important figure in the paper. It shows: (a) within-group distance distributions proving COVID cells are more spread out; (b) displacement test proving centroids are not shifted; (c) donor-level dispersion showing the between-donor heterogeneity mechanism; (d) replication cohort dispersion confirming portability.

11.3 Figure 5 — Transitional compartment

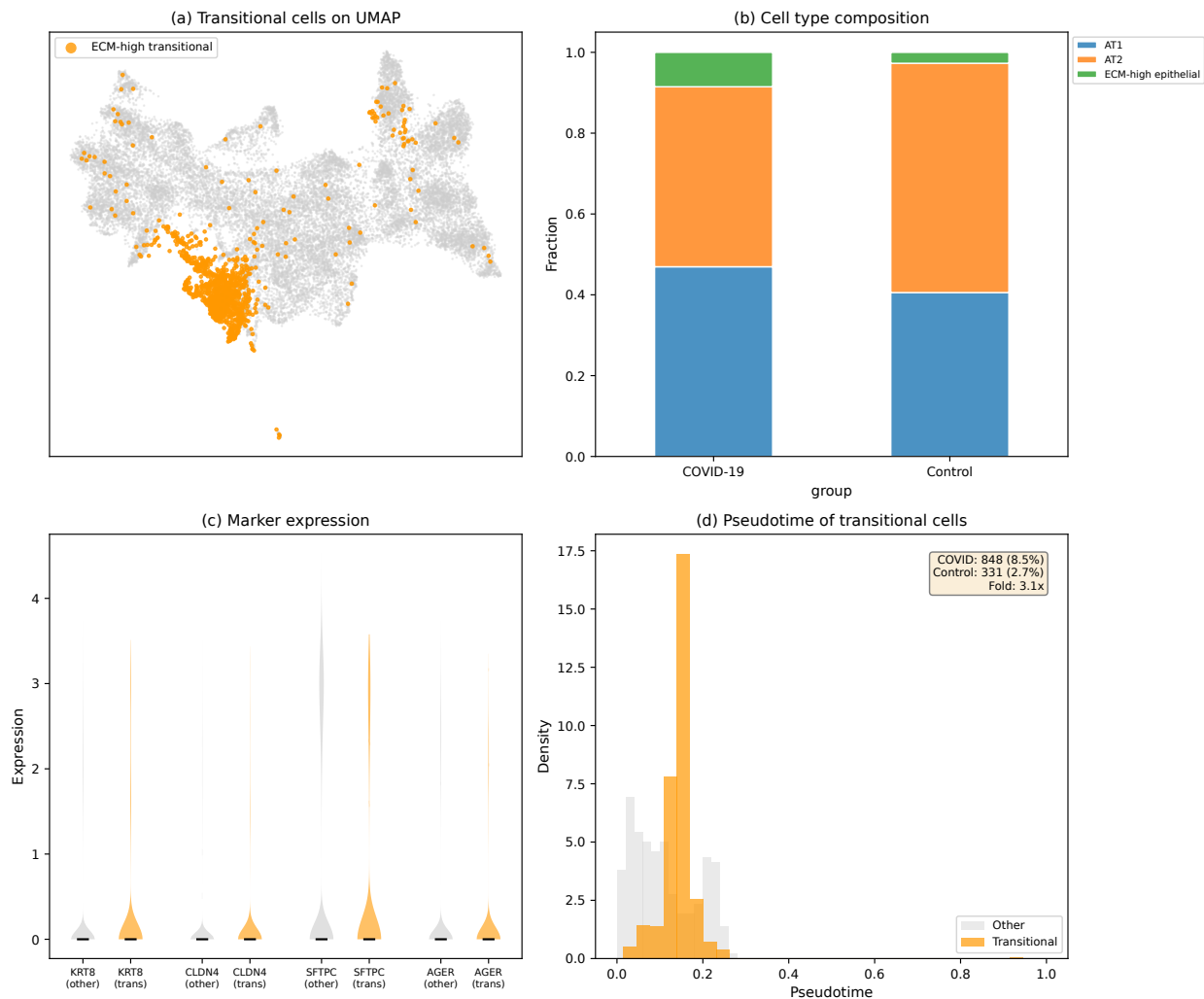


Figure 8: Transitional cell analysis: UMAP highlight, composition bar, markers, pseudotime density.

Transitional cells sit between AT2 and AT1 on the UMAP (intermediate pseudotime), express KRT8/CLDN4, and are $3.1\times$ enriched in COVID. But remember: this enrichment does not replicate cross-cohort using marker thresholds.

12 Part 12 — Summary of key tables

12.1 Table 2: Three pre-registered tests

Test	Statistic	p -value	Effect size	Outcome
Displacement (MW)	5.21×10^7	1.0	$r = 0.14$ (wrong dir.)	Fails
Dispersion (Levene)	276.5	$< 10^{-30}$	ratio $2.3\times$	Supported
Pseudotime (permutation)	—	0.001	MPD $+0.048$	Supported

12.2 State score benchmarks

Score	$\rho(\text{pseudotime})$	MW p	AUC (transitional)
AT2 identity	-0.693	1.7×10^{-25}	0.323
AT1 identity	$+0.687$	5.7×10^{-5}	0.375
Transitional	$+0.282$	1.3×10^{-20}	0.441
Injury composite	$+0.276$	3.8×10^{-70}	0.630
Coherence loss	$+0.412$	1.1×10^{-72}	0.972
Repair failure	$+0.265$	1.3×10^{-55}	0.730

12.3 Donor-level model results

Metric	Value
OLS coefficient (is_covid)	$+0.043$
OLS p -value	0.013
OLS 95% CI	$[0.010, 0.076]$
OLS R^2	0.22
Donor-level MW p	0.026
Bootstrap 95% CI for Δmedian	$[-0.013, +0.086]$
n donors	27

13 Part 13 — How the conclusion was assembled

The conclusion is **not** “we found a new COVID effect.” It is:

Lethal COVID-19 produces dispersion-dominated loss of alveolar epithelial state coherence rather than a coherent directional shift. Cells scatter across heterogeneous injury states while maintaining approximately the same average distance from healthy. The dispersion signal is the strongest, most portable finding — it replicates across 35 independent datasets and 618 donors. Donor-level trajectory direction also replicates. Cell-level displacement does not. Marker-threshold cell-state definitions are atlas-specific.

The reason partial results (2/3 tests succeeding) are more informative than all tests succeeding: the failing test *rules out a specific alternative mechanism*. Having both displacement fail and dispersion succeed tells you the geometry is heterogeneous fan-out, not coherent shift.

New in v3

v3 additions to the conclusion:

1. The cell-level dispersion arises from between-donor heterogeneity: each COVID donor occupies a different region of injury space, collectively producing the fan-out.
2. Continuous state scores (especially coherence loss, $AUC = 0.972$) transfer cross-cohort where marker thresholds fail.
3. The fan-out is driven by divergent activation of senescence, oxidative stress, and NF- κ B programs.
4. Donor-level OLS regression confirms the pseudotime shift at $p = 0.013$ with 22% variance explained.

14 Part 14 — How to reproduce everything

```
# Set up environment
cd sb_class_project
python -m venv env && source env/bin/activate
pip install -r requirements.txt

# Primary pipeline
python scripts/run_pipeline.py --step all

# v3 analyses (donor models, state scores, mechanisms)
python scripts/run_v3_analyses.py

# All ten ablations
bash scripts/ablations/rerun_all.sh

# Independent replication
python scripts/replication/fetch_replication.py
python scripts/replication/analyze_replication.py

# Generate all v3 figures
python scripts/generate_v3_figures.py

# Compile paper
cd manuscript
```

pdflatex paper_v3.tex && pdflatex paper_v3.tex
pdflatex supplementary_methods_v3.tex && pdflatex supplementary_methods_v3.tex

15 Part 15 — Limitations (honest accounting)

1. **Cross-sectional, post-mortem, lethal cases only.** Nothing here tells you about moderate COVID.
2. **27 donors (7 vs 20)** is a small sample. The bootstrap CI for pseudotime difference marginally includes zero. Cross-cohort replication (618 donors) addresses this partially.
3. **Pseudotime is ordering, not elapsed time.** No claim about actual hours or stages.
4. **Replication gene panel too narrow** (84 genes) for a new manifold. Full-transcriptome replication is still missing.
5. **Fine program ordering is hypothesis-generating only.** It inverts under alternative gene lists (Ablation 8).
6. **DATP marker-threshold enrichment retracted** as a portable statistic.
7. **Post-mortem effects.** Peri-mortem hypoxia may influence stress and apoptosis programs.
8. **Between-donor heterogeneity** could partially reflect post-mortem interval differences, not just disease biology.

16 Part 16 — What to tell a friend

To a biologist: In lethal COVID-19, alveolar epithelial cells don't all move to one damaged state. They fan out along multiple axes of injury, driven by divergent activation of senescence, oxidative stress, and inflammatory programs. This "state coherence loss" signature replicates across 618 donors in 35 independent datasets. Marker-threshold DATP assays don't transfer cross-cohort, but continuous manifold-based scores do.

To a computer scientist: We pre-registered three geometric tests with mutually exclusive mechanistic interpretations, ran ten ablations, fixed a 1-line Harmony bug that changed the effect size $2.6\times$, replicated in a 618-donor held-out atlas, and added donor-level OLS/bootstrap modeling to address the $n = 27$ problem. The methodological takeaway: dispersion beats centroid displacement for heterogeneous disease states, donor-level pseudobulk beats cell-level tests, and continuous state scores beat marker thresholds for cross-cohort portability.

To yourself: The lung epithelium in lethal COVID doesn't collapse by going somewhere new — it collapses by spreading out. The cleanest way to see that is variance, not distance. And the spread comes from donors being different from each other, not from individual donors being internally messy.